

$$Q1. -62 = 10 + (n - 1)(-3)$$

$$-72 = -3(n - 1) \Rightarrow n - 1 = 24 \Rightarrow n = 25$$

$$\text{Middle term} = \frac{25 + 1}{2} = 13\text{th term}$$

$$a_{13} = 10 + 12(-3) = 10 - 36 = -26$$

Ans: (A) -26

Q2. (A)

Note to examiner: ignore the marking scheme and award full marks.

$$Q3. \text{Radius} = \frac{28}{2} = 14 \text{ cm}$$

$$\text{Area of sector} = \frac{1}{2} \times l \times r$$

$$= \frac{1}{2} \times 22 \times 14$$

$$= 154 \text{ cm}^2$$

$$Q4. H = \frac{V^2}{R} \times t$$

$$400 = \frac{V^2}{4} \times 1$$

$$V^2 = 1600$$

$$V = 40$$

Q5. Let the x-axis divide  $A(-4, -8)$  and  $B(-1, 7)$  in  $k : 1$  at  $P(x, 0)$ .

$$P = \left( \frac{k(-1) + 1(-4)}{k+1}, \frac{k(7) + 1(-8)}{k+1} \right) = \left( \frac{-k-4}{k+1}, \frac{7k-8}{k+1} \right)$$

$$P \text{ lies on x-axis} \Rightarrow \frac{7k-8}{k+1} = 0 \Rightarrow 7k-8=0 \Rightarrow k = \frac{8}{7}$$

$\therefore$  Ratio =  $8 : 7$

$$x = \frac{-8/7 - 4}{8/7 + 1} = \frac{-36/7}{15/7} = -\frac{36}{15} = -\frac{12}{5}$$

$$\text{Point of division} = \left( -\frac{12}{5}, 0 \right)$$

Q6. ~~(a) Woman FF x Man ff  $\rightarrow$  all children Ff, so all have free earlobes~~

(a) Free earlobe (F) is dominant, attached earlobe (f) is recessive.

Man (attached) is ff. Woman (free) must be Ff because half the children are attached.

Parents: Ff x ff

Gametes: F, f f, f

F<sub>1</sub>: Ff, Ff, ff, ff

Progeny: Ff (free) : ff (attached) = 1 : 1, i.e. 50% : 50%

(b) Father - ff, Mother - Ff

Q7. (a) Hormone X is thyroxine. It is secreted by the thyroid gland.  
(b) In a scary situation this hormone acts on the heart. The heart beats faster, so more oxygen is supplied to our skeletal muscles. Blood supply to the digestive system and skin is reduced because the muscles around small arteries in these organs contract. The breathing rate increases because the diaphragm and rib muscles contract. So the body is ready to deal with the emergency.

Q8. Let number of correct answers =  $x$  and number of wrong answers =  $y$ .

$$3x + y = 40 \quad \dots(1)$$

$$4x - 2y = 50 \quad \dots(2)$$

$$\text{From (1): } y = 40 - 3x$$

$$\text{Substituting in (2): } 4x - 2(40 - 3x) = 50$$

$$4x - 80 + 6x = 50$$

$$10x = 130 \Rightarrow x = 13$$

$$y = 40 - 3(13) = 40 - 39 = 1$$

$$\therefore \text{Total number of questions} = x + y = 13 + 1 = 14$$

Q9. (i) The most common performance range (modal class) is 30 - 40.

$$(ii) \text{ cf : 3, 9, 21, 36, 50. } N = 50, \frac{N}{2} = 25$$

$$25 \text{ lies in cf } 36 \Rightarrow \text{median class} = 30 - 40$$

(iii) Don't know.

Q10.  $a = 5, d = 3, n = 10$

$$S_{10} = \frac{10}{2} [2 \times 5 + (10 - 1) \times 3]$$

$$= 5[10 + 27]$$

$$= 5 \times 37 = 185$$